

Chapter: LeetCode 2045 - Second Minimum Time to Reach Destination

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Visualized Logic & Notes

1. Objective

filhaal ham sirf second shortest path nikaalne me interested he

2. Graph Traversal Strategy

bfs cause equal weight.

Implementation: `queue<pair<int,int>> q;` (isme node aur distance store krenge).

3. Path to Time Conversion

timing will be handled at last using change simulation.

now we need to convert second shortest distance to second shortest time.

ill simulate edge by edge.

4. Traffic Signal Simulation

- `interval % 2 == 1` : matlab red he wait krna padega till next interval.
- `currt += time` : green he to chal jaao , red he to wait krne ke baad jaao.

C++ Source Code Implementation

```
class Solution {
public:
    int secondMinimum(int n, vector<vector<int>>& edges, int time, int change) {
        vector<vector<int>> adj(n+1);
        for(auto &it:edges){
            int u=it[0];
            int v=it[1];

            adj[u].push_back(v);
            adj[v].push_back(u);
        }

        vector<int> dist1(n,INT_MAX);
        vector<int> dist2(n,INT_MAX); //for second shortest path
```

```

//filhaal ham sirf second shortest path nikaalne me interested he
//bfs cause equal weight
//timing will be handled at last using change simulation

queue<pair<int,int>> q; //isme node aur distance store krengae
q.push({1,0});
dist1[1]=0;

while(!q.empty()){
    int currnode=q.front().first;
    int currdiss=q.front().second;
    q.pop();

    for(auto& it:adj[currnode]){
        int newnode=it;
        int newdiss=currdiss+1;

        if(newdiss<dist1[newnode]){
            dist2[newnode]=dist1[newnode];
            dist1[newnode]=newdiss;
            q.push({newnode,newdiss});
        }

        else if(newdiss>dist1[newnode] && newdiss<dist2[newnode]){
            dist2[newnode]=newdiss;
            q.push({newnode,newdiss});
        }
    }
}

//now we need to convert second shortest distance to second
shortest time

//ill simulate edge by edge
int edgess=dist2[n];
int currt=0;
for(int i=0;i<edgess;i++){
    int interval=currt/change;
    if(interval%2==1){
        //matlab red he wait krna padega till next interval
        currt=(interval+1)*change;
    }

    currt+=time; //green he to chal jaao , red he to wait krne ke
    baad jaao

```

```
    }  
    return currt;  
}  
};
```